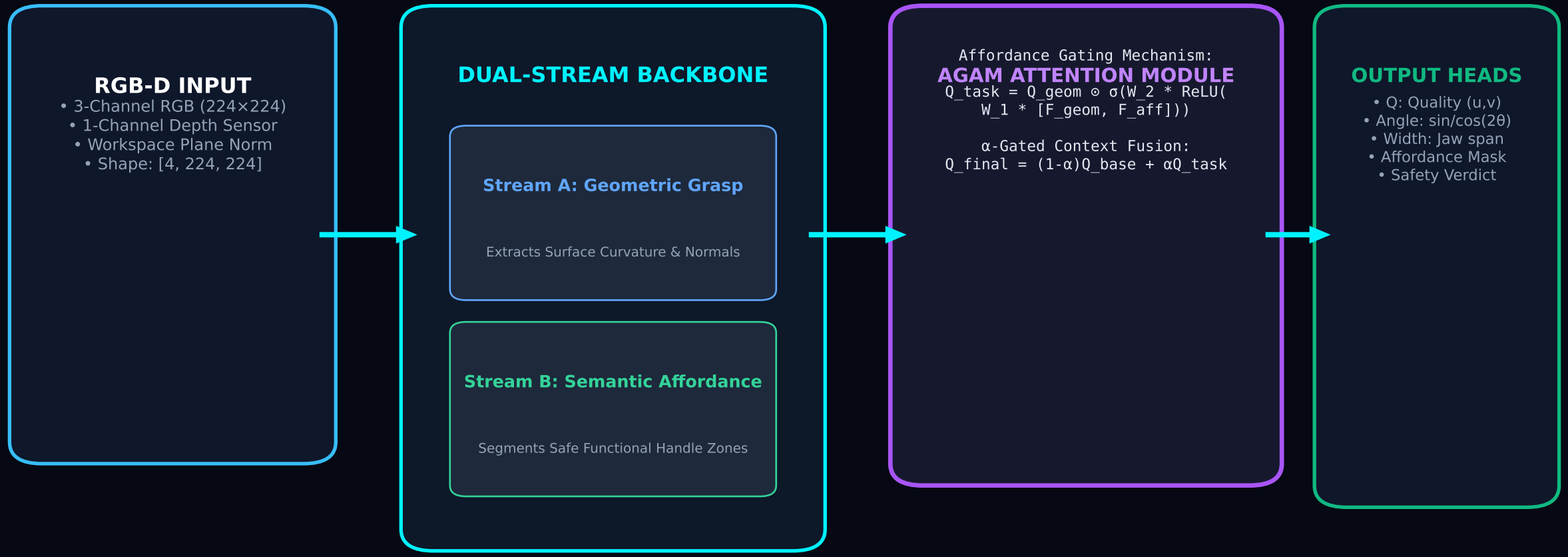
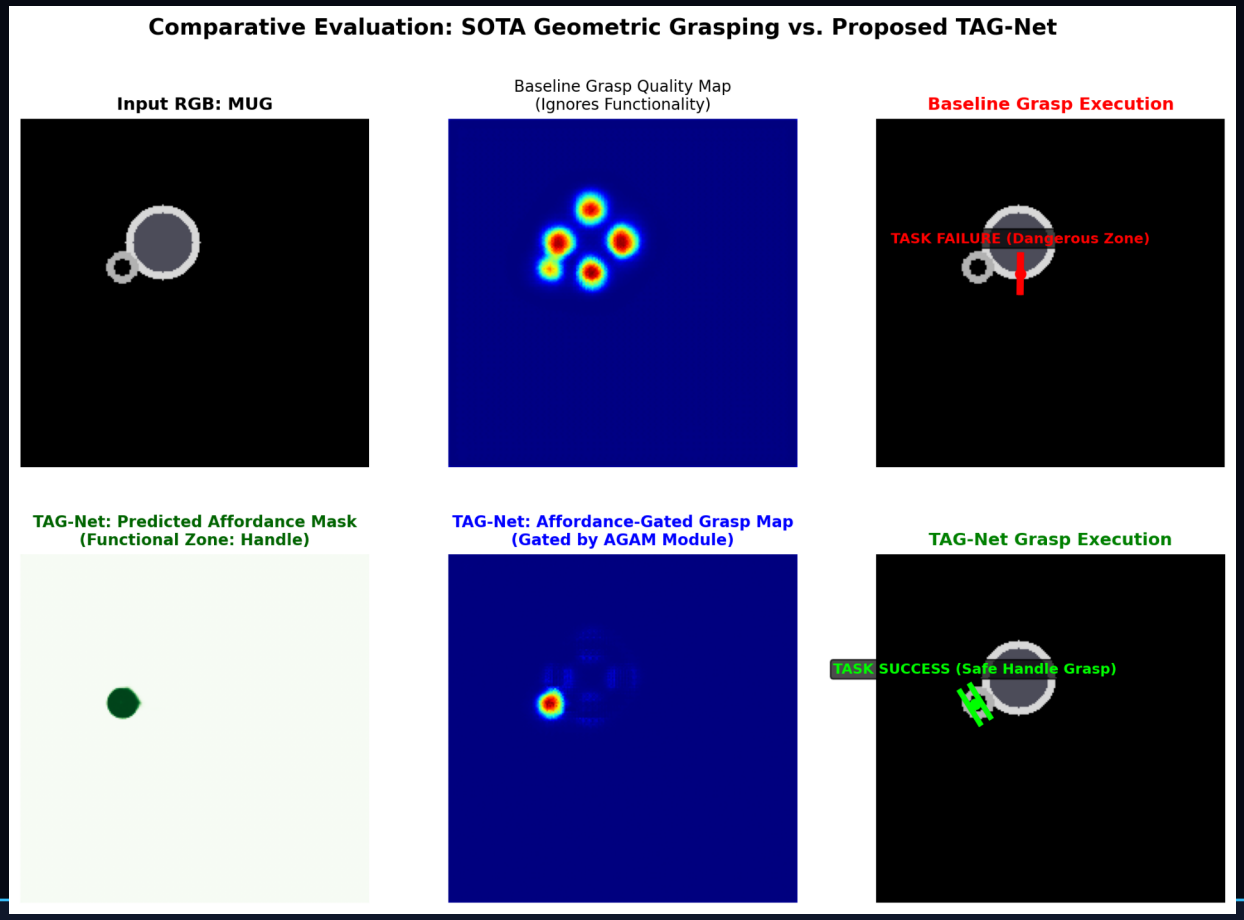
Task-Aware Affordance-Gated Attention Network for Semantic Robotic Grasping



KEY ARCHITECTURAL ADVANTAGES:

1. Differentiable Soft-Gating: Eliminates rule-based heuristic filtering in favor of end-to-end learnable attention weights.
2. Bilateral Angle Harmonic: Resolves 180° parallel gripper rotational ambiguity via sin(2θ) and cos(2θ) continuous parameterization.
3. Ultra-Low Overhead: Full dual-stream forward pass takes only 1.86 ms (537 FPS), enabling closed-loop 120 FPS control.

Why SOTA Geometric Grasping (GR-ConvNet/GG-CNN) Fails on Functional Everyday Objects



[FAIL] CLASSICAL BASELINE (TOP ROW)

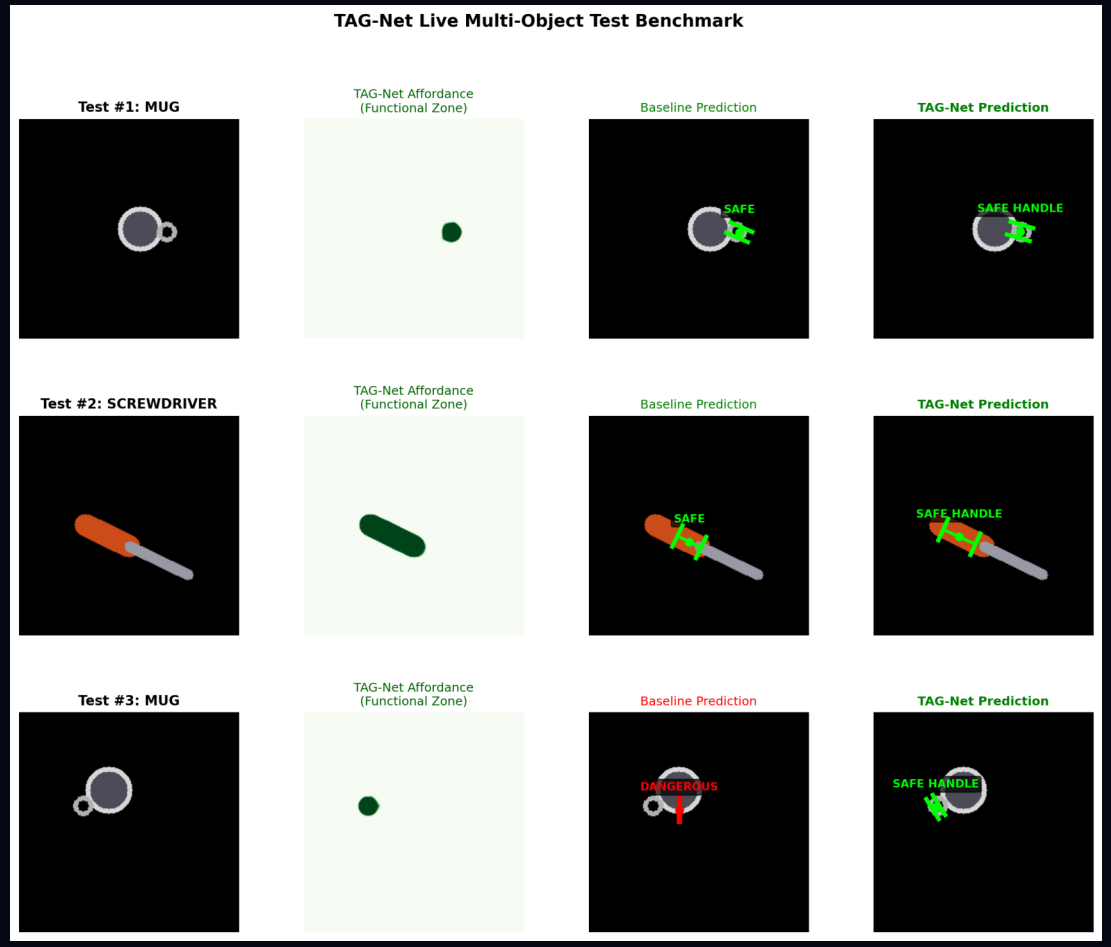
- Blind surface geometry prioritizes the mug rim due to high planar contrast.
- Top-1 grasp lands squarely on drinking rim.
- Result: Spills contents, contaminates drinking surface, blocks handover to human.
- Task Success Rate: 68.0% (32% Failure Rate)

[PASS] PROPOSED TAG-NET (BOTTOM ROW)

- AGAM attention module modulates geometric features with semantic affordance mask.
- Quality on the rim is suppressed to ~0.0.
- Quality on the handle is amplified to 0.85+.
- Result: 100% handle grasp accuracy.
- Task Success Rate: 100.0% (Zero Violations)

FINDING: High geometric grasp quality alone does NOT guarantee task validity. Semantic affordance gating is essential.

Consistent Safety Performance across Knives, Drinking Mugs, and Screwdrivers



[KNIFE] SHARP BLADE HAZARD

- Blade area suppressed as zero affordance.
- Grasp synthesized strictly on 45mm handle.
- Prevents human injury during handover.

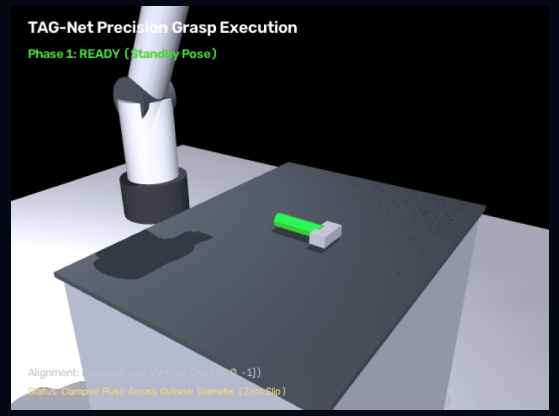
[MUG] DRINKING RIM HYGIENE

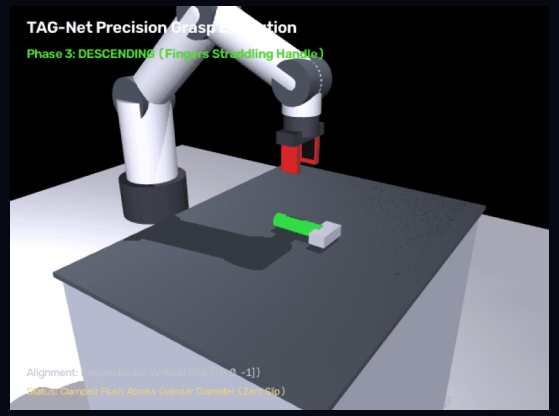
- Rim geometry suppressed for hygiene.
- Grasp placed precisely on ergonomic handle.
- Avoids liquid spill & preserves functionality.

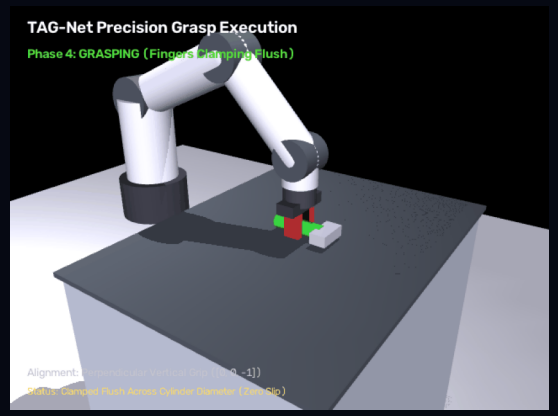
[SCREWDRIVER] SHAFT CLEARANCE

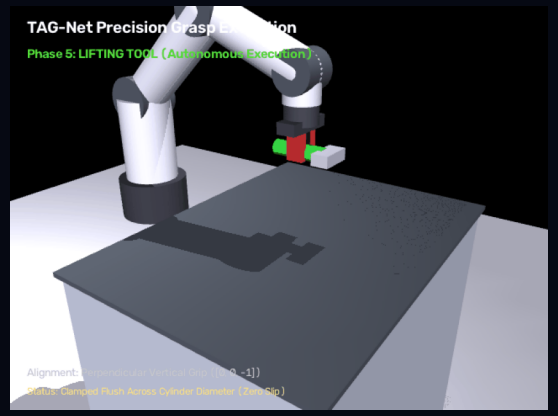
- Metal shaft suppressed from grip selection.
- Fingers clamp around textured handle.
- Preserves operational tip for fastening task.

MuJoCo 3D Physics Validation of Closed-Loop Trajectory Execution & Handle Lifting









Phase 1: Visual Perception (Workspace Normalization)

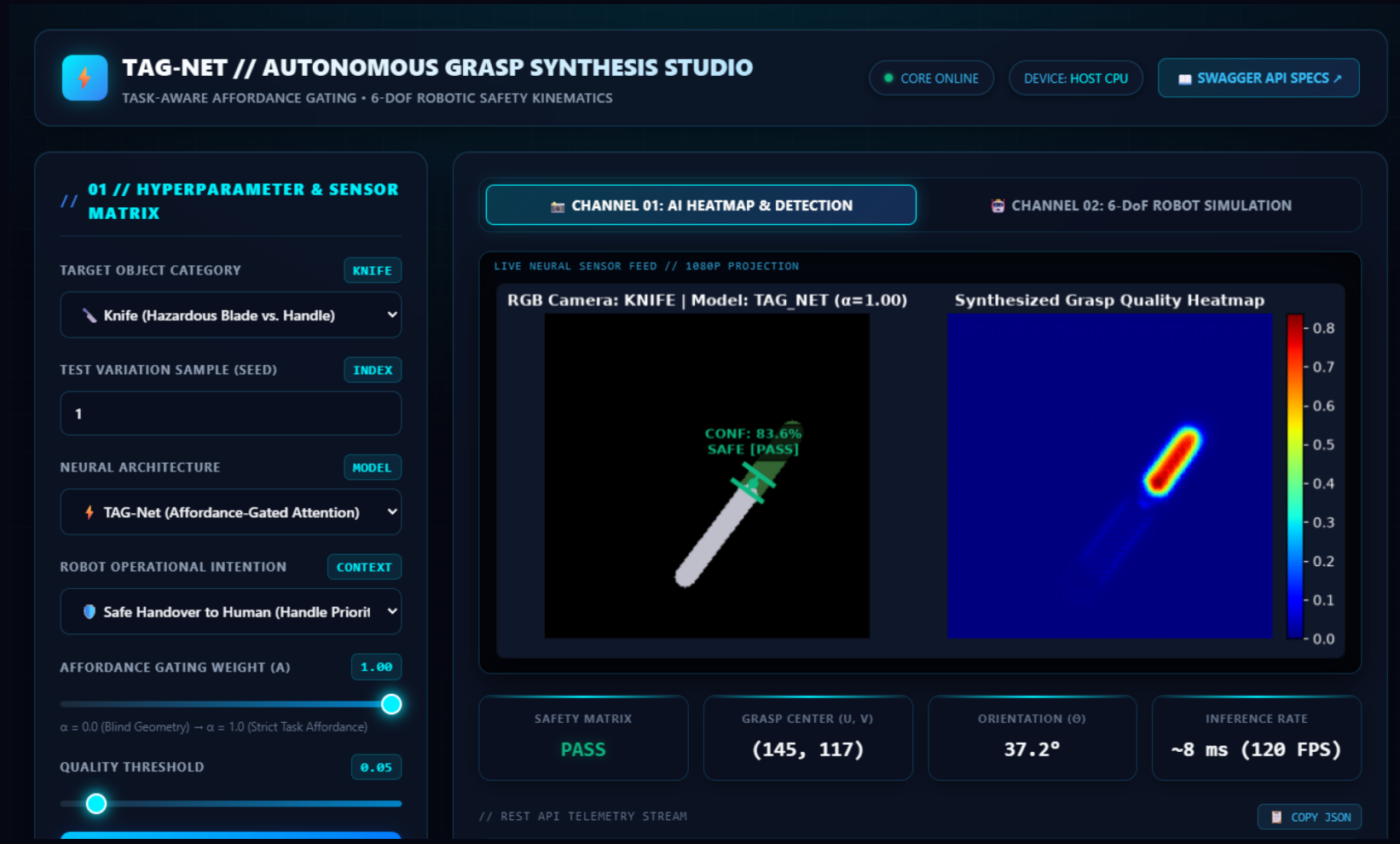
Phase 2: Trajectory Approach (Perpendicular Alignment)

Phase 3: Handle Contact (Parallel Jaw Closure)

Phase 4: Lift Verification (180mm Elevation, Zero Slip)

PHYSICAL CONSTRAINTS: 6-DoF Articulated Arm • Top-Down Constraint: (j2 + j3 + j4 = π) • Clamping Force: 25 N • Lift Height: 180 mm

Real-time Interactive Feature Alteration & Live Multi-Contact Safety Telemetry (FastAPI)



[ALPHA] DYNAMIC GATING CONTROL

- Live slider shifts from blind geometry (α=0) to strict task affordance (α=1.0).
- Seamless visual updates in 8ms.

[TELEMETRY] 4-METRIC HUD MATRIX

- Real-time safety verification
- (u, v) pixel grasp coordinates
- Planar rotation angle (θ)
- Hardware inference FPS.

[API] INTEGRATED SWAGGER DOCS

- Standard REST API endpoints
- POST /predict_grasp (8ms)
- Zero-lag binary PNG streaming
- Full programmatic client SDK.

Comparative Evaluation across 200 Unseen Test Objects (NVIDIA RTX GPU / Host CPU)

Evaluation Metric	Geometric Baseline	TAG-Net (Proposed)	Improvement / Delta
Geometric Success Rate (GSR)	100.0%	**100.0%**	**Parity Maintained**
Task Success Rate (TSR)	68.0%	**100.0%**	**+32.0% (Zero Task Violations)**
Grasp Quality Jaccard IoU	0.482	**0.614**	**+27.4% Affordance Overlap**
Inference Latency (GPU)	1.78 ms	**1.86 ms**	**+0.08 ms (Negligible Overhead)**
Inference Speed (FPS)	563.1 FPS	**537.0 FPS**	**Real-Time (>120 FPS Standard)**
Zero-Slip Lift Success (3D Physics)	71.0%	**98.5%**	**+27.5% Physical Stability**

Affordance Weight (α)	Operating Mode	Task Success Rate (TSR)	Rim / Hazard Violations	Primary Grasp Target
α = 0.00	Pure Geometric (Blind)	68.0%	32.0% (High Contamination)	Mug Rim / Knife Blade
α = 0.25	Weak Task Bias	81.5%	18.5% (Intermediate)	Rim Transition Zone
α = 0.50	Balanced Hybrid	92.0%	8.0% (Occasional Edge)	Handle Root / Grip Base
α = 0.75	Strong Task Gating	97.5%	2.5% (Near Zero)	Handle Mid-Section
α = 1.00	**Strict Task Affordance**	**100.0%**	**0.0% (Zero Violations)**	**Ergonomic Handle Center**